

Chipmoji

An emoji guide for chip development commits.

GIT 16

 **:tada:**
Initial commit or project bootstrap.

 **:bug:**
Fix functional bugs in RTL, verification, Python, scripts, or flow code.

 **:ambulance:**
Fix urgent breakage: failing regression, broken CI, lab blocker, release blocker, or tapeout blocker.

 **:sparkles:**
Add a complete new feature.

 **:construction:**
Commit incomplete work intentionally.

 **:boom:**
Breaking interface, register map, protocol, API, timing contract, or CLI change.

 **:recycle:**
Refactor without intended behavior change.

 **:art:**
Improve structure, formatting, naming, or readability without behavior change.

 **:fire:**
Remove obsolete code, files, stale scripts, or generated artifacts.

 **:coffin:**
Remove dead signals, unreachable FSM states, unused functions, or obsolete helpers.

 **:wastebasket:**
Deprecate code or behavior that should be removed later.

 **:truck:**
Move or rename files, modules, packages, paths, or resources.

 **:rewind:**
Revert a previous change.

 **:twisted_rightwards_arrows:**
Merge branches. Do not use for muxing, routing, or arbitration.

 **:see_no_evil:**
Add or update `.gitignore`.

 **:bookmark:**
Release or version tag.

RTL 12

 **:bridge_at_night:**
Clock-domain crossing, async boundary, synchronizer, handshake bridge, or reset-domain crossing.

 **:rightwards_pushing_hand:**
Valid/ready pushback, stall propagation, skid buffers, flow control, or downstream backpressure.

 **:electric_plug:**
Wire ports, buses, protocols, register interfaces, AXI/APB/TileLink/Wishbone connections, or top-level integration.

 **:abacus:**
Arithmetic, bit-width, fixed-point, saturation, rounding, accumulator, multiplier, divider, or datapath logic.

 **:vertical_traffic_light:**
Arbitration, priority, scheduling, traffic shaping, QoS, credits, or grant logic.

 **:compass:**
Address maps, memory maps, topology, routing tables, NoC coordinates, or integration direction.

 **:dna:**
Generics, parameters, SKUs, feature variants, configurable widths/depths, or generated RTL variants.

 **:fire_extinguisher:**
Fix X-prop, glitch risk, unsafe reset behavior, metastability risk, race, or silicon-escape-class hazard.

 **:magnet:**
Board-level work, IO, pins, PHYs, lab-only fixes, FPGA bring-up, or constraints tied to real hardware.

 **:satellite_antenna:**
PCIe, Ethernet, DDR, SerDes, SPI, I2C, UART, or other external protocol/link behavior.

 **:safety_pin:**
Small localized hardware workaround or mitigation for a known issue.

 **:screwdriver:**
Small mechanical RTL adjustment too narrow for a bigger category.

TIMING 2

 **:scissors:**
Reduce combinational depth, retime logic, split paths, or improve Fmax.

 **:stopwatch:**
Benchmark instrumentation, cycle counting, latency measurement, or timing measurement infrastructure.

PPA 4

 **:zap:**
Improve speed, throughput, latency, benchmark result, or runtime.

 **:chart_with_downwards_trend:**
Reduce LUTs, FFs, gates, SRAM, BRAM, DSPs, cells, or general resource usage.

 **:chart_with_upwards_trend:**
Accept area, power, or resource increase intentionally for feature, timing, verification, or performance.

 **:battery:**
Clock gating, power gating, idle behavior, low-power mode, activity reduction, or power optimization.

VERIFICATION 7

 **:white_check_mark:**
Add, update, or fix passing tests, regressions, assertions, formal checks, or checkers.

 **:test_tube:**
Add a failing test or reproducer for a known bug.

 **:rotating_light:**
Fix compiler, linter, CDC, synthesis, simulation, or timing warnings.

 **:game_die:**
Randomized testing, constrained-random stimulus, fuzzing, seeds, or randomized regressions.

 **:microscope:**
Add probes, traces, counters, waveform hooks, debug CSRs, or observability.

 **:safety_vest:**
Add validation, guards, illegal-state checks, protocol checks, or defensive checks.

 **:seedling:**
Add seed files, sample inputs, golden data, or deterministic regression data.

PYTHON 5

 **:snake:**
Python-specific implementation work when no more precise standard Gitmoji applies.

 **:label:**
Add or update Python type hints, dataclasses, TypedDicts, Pydantic models, SV typedefs, enums, or structs.

 **:goal_net:**
Improve exceptions, parser errors, failure handling, or user-facing error messages.

 **:monocle_face:**
Data exploration, report inspection, one-off analysis, or investigation scripts.

 **:thread:**
Threading, multiprocessing, async, job queues, or concurrent report processing.

SCRIPTS 2

 **:hammer:**
Build, sim, synth, lint, formal, regression, packaging, or helper scripts.

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:wrench:

Tool config, environment config, CLI config, YAML/TOML/JSON config, synthesis/sim flags, or parameters.

CI 2



:construction_worker:

CI pipelines, GitHub Actions, build matrix, scheduled jobs, or regression automation.



:green_heart:

Fix broken CI, regression automation, build matrix, or tool-cache issue.

DOCS 4



:memo:

Add or update docs, specs, register maps, integration notes, or README content.



:bulb:

Add or update source comments explaining hardware behavior, assumptions, or constraints.



:pencil2:

Fix typos in docs, comments, specs, or messages.



:page_facing_up:

Add or update license.

DEPENDENCIES 6



:arrow_up:

Upgrade dependency, vendor IP, tool version, package, or submodule.



:arrow_down:

Downgrade dependency, vendor IP, tool version, package, or submodule.



:pushpin:

Pin exact versions of tools, IPs, packages, lockfiles, containers, or submodules.



:heavy_plus_sign:

Add a dependency, package, included IP, submodule, or tool requirement.



:heavy_minus_sign:

Remove a dependency, package, included IP, submodule, or tool requirement.



:alien:

Update code due to external API, protocol, vendor IP, tool output, or report-format changes.

INFRASTRUCTURE 2



:bricks:

Hardware-flow infrastructure: runners, artifact storage, caches, job orchestration, or shared tool setup.



:package:

Add or update generated bundles, packaged IP, release artifacts, reports, or bitstreams.